

NAKUL BUKKAWAR

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SUMMARY

Backend and full-stack developer with 3.5+ years building production microservices (Quarkus, Spring Boot, Angular, GCP/AWS/Azure) for enterprise clients including BMW and HDFC Bank. Hands-on applied ML experience engineering feature pipelines (Pandas/NumPy) for a production demand-forecasting model and integrating it into serverless inference services. Actively building GenAI/agentic engineering skills: built a working RAG pipeline on PostgreSQL/pgvector, implemented an MCP server, prototyped an LLM-first automated-mapping pipeline with deterministic fallback using the Anthropic API, and completed coursework on agentic AI systems. Strong foundation in REST/API design, event-driven systems, and cloud infrastructure.

Core Competencies: Microservices Architecture | Cloud-Native Development | REST API Design | Database Optimization | Event-Driven Systems | CI/CD Automation | Reactive Programming | LLM API Integration | RAG | Prompt Engineering | Full-Stack Development | Agile/Scrum

CORE SKILLS

Languages: Java, Python, TypeScript, JavaScript, Bash

Backend & APIs: Quarkus, Spring Boot, Node.js, FastAPI, REST API design, Mutiny/reactive programming, RxJS

Frontend: Angular (RxJS, NgRx), React (Redux, Hooks), HTML5/CSS3

Cloud Platforms: Google Cloud Platform (Cloud Functions, API Gateway, GCS, Cloud Filestore), AWS (Lambda, S3, EC2, RDS), Microsoft Azure (Functions, App Service, Blob Storage) — AWS & GCP certified

Data & Messaging: PostgreSQL, MongoDB, MySQL, Flyway migrations, RabbitMQ, Apache Kafka

DevOps: Docker, Docker Compose, Kubernetes, GitHub Actions, GitLab CI/CD, Jenkins, Git

Applied ML (hands-on): Feature engineering (Pandas/NumPy) for production forecasting models — categorical encoding, seasonality/trend feature construction; ML model inference integration in production services

GenAI / LLM (hands-on): Anthropic API (Claude) integration, prompt engineering, RAG (retrieval-augmented generation) with PostgreSQL/pgvector, MCP (Model Context Protocol) server implementation, LangGraph (agent workflows, in progress)

GenAI / Agentic (actively learning): Multi-agent orchestration, prompt evaluation (ragas), agentic system design patterns

Testing: JUnit, Jasmine, Karma, Postman, Swagger

GENAI PROJECTS

Retrieval-Augmented Generation (RAG) Pipeline

- Built a working RAG system using PostgreSQL with pgvector for embedding storage and retrieval, paired with the Anthropic API for generation over a personal document set — covering chunking, embedding, similarity search, and grounded response generation.

MCP (Model Context Protocol) Server

- Implemented a functional MCP server exposing a custom tool to an LLM client, following the MCP specification — running locally end-to-end.

LangGraph Agent Workflow (in progress)

- Built a working multi-node LangGraph workflow as a foundation for agentic behavior; extending toward full planner → tool-call → response orchestration.

PROFESSIONAL EXPERIENCE

Full Stack Developer | BMW TechWorks India, Pune | June 2025 – Present

- Building the CO2MOS Importer platform (Angular + Quarkus/Java microservices) for vehicle component-to-sustainability-data mapping, used across BMW vehicle programmes for Product Carbon Footprint (PCF) assessment.
- Designed and optimized a rule-based automated matching algorithm that scores component-to-sustainability-data pairs by weighted similarity, replacing manual SME lookup for the majority of records and reducing confidence-scoring bugs by moving from weighted-average to max-based sub-component scoring.
- Eliminated N+1 query patterns by introducing bulk-fetch REST endpoints, measurably reducing per-vehicle load times.
- Prototyped an LLM-first mapping pipeline using the Anthropic API (Claude) with deterministic rule-based fallback, evaluating GenAI-assisted matching as a single-step agentic pattern against the existing scoring engine.
- Implemented a two-stage human-in-the-loop review workflow (assigned → confirmed states, SME Review tab) to keep automated suggestions auditable before they enter production data.

- Own Flyway-based DB migrations, Docker Compose environments, and CI/CD deploy automation (GitHub Actions) across frontend and backend repos.

Full Stack Developer | IBM, Pune | *June 2022 – May 2025*

Fabric.ai Data Platform for Bestseller India (Jan 2025 – May 2025)

- Led end-to-end integration of the Fabric.ai platform into enterprise infrastructure, designing the cloud-native data ingestion and real-time processing pipeline feeding a machine-learning demand-forecasting model used to predict likely commercial performance and optimal launch season for new apparel designs.
- Engineered the feature processing pipeline (Pandas, NumPy) that transformed raw garment metadata and multi-year historical sales trends into model-ready inputs, including encoding categorical design attributes and constructing seasonality/trend features from historical data — model training itself was owned by the data science team.
- Integrated the trained model into production by calling its prediction endpoint from serverless Python Cloud Functions, handling payload construction, error handling for missing/unseen attribute values, and post-processing raw scores into designer-facing launch-window recommendations.
- Implemented a secure GCP API Gateway centralizing API management across 15+ microservices, and designed a hybrid storage architecture (Cloud Filestore + GCS) balancing retrieval speed against infrastructure cost for training/inference data.

FASTAG Migration for HDFC Bank (Jan 2024 – Dec 2024)

- Spearheaded migration of a legacy C# monolith processing 2M+ daily transactions to a Java/Quarkus microservices architecture, with zero downtime and full backward compatibility; engineered the backend with comprehensive load testing, improving response times and throughput.
- Led PostgreSQL schema redesign, data validation, and performance tuning as part of the migration, with rollback procedures ensuring zero data loss; collaborated with frontend engineers on Flutter Flow interfaces.
- Implemented RabbitMQ for asynchronous inter-service communication and redesigned API architecture with error handling and circuit breakers, improving resilience, uptime, and incident response time.

Common Test Platform for AT&T (Aug 2023 – Dec 2023)

- Transformed manual testing into an automated framework (REST Assured, JUnit, Drools), raising test coverage from 45% to 85% and increasing release velocity.
- Automated provisioning workflows using Bash and Jenkins, cutting recurring monthly manual effort, and established CI/CD best practices and regression suites to improve pre-production defect capture.

API Development & Support for Horizon BCBS (Sep 2022 – Jul 2023)

- Managed mission-critical healthcare APIs processing high daily request volumes; reduced production failures significantly through systematic debugging, refactoring, and enhanced error handling.
- Optimized API performance through caching, query optimization, load balancing, and resource pooling; built scalable microservices (Spring Boot, Apache Kafka, MongoDB) supporting significant traffic growth without infrastructure expansion.
- Implemented real-time monitoring dashboards using Splunk for proactive issue resolution, and collaborated with frontend teams on Angular/React integration to reduce client-side errors.

EDUCATION

Bachelor of Engineering — Electronics & Communication (Minor: Computer Science)

Shri Ramdeobaba College of Engineering and Management, Nagpur, Maharashtra | May 2022 | CGPA: 8.35/10

ACADEMIC PROJECTS

Smart Saline System with Message Alert

- IoT-based healthcare monitoring system using Flash, MQTT protocol, Arduino microcontroller, and MySQL for real-time patient vital tracking and automated alerts.

Smart Door Lock with ML Authentication

- Computer-vision security system using TensorFlow, OpenCV, and Python for facial-recognition-based access control.

CERTIFICATIONS & COURSES

- AWS Certified Developer – Associate
- AWS Certified Cloud Practitioner
- Google Cloud Digital Leader
- Microsoft Certified: Azure Fundamentals
- Agentic AI and AI Agents: A Primer for Leaders